

Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the generalized Doppler shift is derived in three dimensions directly from the algebraic properties of the bivector field and the convective derivative operator D .

By representing the electromagnetic wave as a $Cl(4, 1, 1)$ field bivector F_{wave} and solving the vacuum field equation $DF_{\text{wave}} = 0$, we obtain the convective dispersion relation. Combining this with the physical field-mass inertia of the moving wave packet yields the exact Doppler shift.

Section 8: 3D Convective Derivation in the $Cl(4, 1, 1)$ Algebra System

8.1 The $Cl(4, 1, 1)$ Convective Wave Equation The spacetime derivative operator in $Cl(4, 1, 1)$ is defined as [2.1]:

$$D = \nabla + e_4 \frac{1}{c} \left(\frac{\partial}{\partial t} - \mathbf{u} \cdot \nabla \right)$$

where: ${}^* \nabla = \sum_{j=1}^3 e_j \partial_j$ is the spatial gradient (grade 1). ${}^* \mathbf{u} = \sum_{j=1}^3 u_j e_j$ is the constant velocity vector of the moving transmitter relative to the stationary target at the origin. ${}^* e_4$ is the null temporal basis vector ($e_4^2 = 0$) [1.1].

The electromagnetic wave is represented as a Grade-2 field bivector [2.2]:

$$F_{\text{wave}} = E e_4 + c B I_3$$

where E is the spatial electric vector (grade 1), B is the spatial magnetic bivector (grade 2), and $I_3 = e_{123}$ is the spatial pseudo-scalar.

The vacuum field equation is [2.4]:

$$DF_{\text{wave}} = 0$$

8.2 Plane-Wave Solution in $Cl(4, 1, 1)$ We assume a 3D monochromatic plane-wave bivector solution propagating toward the stationary target:

$$F_{\text{wave}}(\mathbf{x}, t) = F_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_{\text{rec}} t)}$$

where: ${}^* F_0 = E_0 e_4 + c B_0 I_3$ is the constant amplitude bivector. ${}^* \mathbf{k}$ is the spatial wave vector. ${}^* \omega_{\text{rec}}$ is the wave frequency registered at the target.

Applying the convective derivative operator D to this plane-wave bivector translates the derivative into a geometric product with an algebraic $Cl(4, 1, 1)$ wave vector K_D :

$$DF_{\text{wave}} = i K_D F_{\text{wave}} = 0 \implies K_D F_0 = 0$$

where the $Cl(4, 1, 1)$ wave vector is:

$$K_D = \mathbf{k} - e_4 \frac{1}{c} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k})$$

8.3 Derivation of the Convective Dispersion Relation The geometric product $K_D F_0 = 0$ is expanded in $Cl(4, 1, 1)$:

$$\left[\mathbf{k} - e_4 \frac{1}{c} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) \right] [E_0 e_4 + c B_0 I_3] = 0$$

$$\mathbf{k} E_0 e_4 + c \mathbf{k} B_0 I_3 - e_4 \frac{1}{c} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) E_0 e_4 - e_4 (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0 I_3 = 0$$

Since $e_4^2 = 0$ [1.1], the third term $e_4 E_0 e_4 = -E_0 e_4^2 = 0$ vanishes.

Using the anticommutation relations of the spatial pseudo-scalar ($e_4 B_0 I_3 = -B_0 I_3 e_4$), the equation simplifies to:

$$\mathbf{k} E_0 e_4 + c \mathbf{k} B_0 I_3 + (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0 I_3 e_4 = 0$$

Grouping the terms with the null temporal vector e_4 :

$$[\mathbf{k} E_0 + (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0 I_3] e_4 + c \mathbf{k} B_0 I_3 = 0$$

Because e_4 and I_3 are linearly independent, the terms inside and outside the bracket must vanish independently: 1. **Solenoidal/Transverse Condition:**

$$c \mathbf{k} B_0 I_3 = 0 \implies \mathbf{k} B_0 = 0$$

2. **Precessional Wave Equation:**

$$\mathbf{k} E_0 + (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0 I_3 = 0$$

Because $\mathbf{k} \cdot E_0 = 0$ (from the divergence-free condition), the geometric product is equal to the outer product, which is the dual of the classical curl: $\mathbf{k} E_0 = \mathbf{k} \wedge E_0 = (\mathbf{k} \times E_0) I_3$. Substituting this into the precessional wave equation:

$$(\mathbf{k} \times E_0) I_3 = -(\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0 I_3$$

Dividing out I_3 yields:

$$\mathbf{k} \times E_0 = -(\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k}) B_0$$

Taking the magnitude of the cross-product (where $\mathbf{k} \cdot \mathbf{E}_0 = 0$):

$$k^2 E_0^2 = (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k})^2 B_0^2$$

Similarly, the dual convective Ampère's law yields:

$$k^2 B_0^2 = \frac{1}{c^4} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k})^2 E_0^2$$

Multiplying these two magnitude equations and dividing out the non-zero amplitudes $E_0^2 B_0^2$:

$$k^4 = \frac{1}{c^4} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k})^4 \implies k^2 = \frac{1}{c^2} (\omega_{\text{rec}} - \mathbf{u} \cdot \mathbf{k})^2$$

This yields the convective dispersion relation:

$$\omega_{\text{rec}} = ck \left(1 + \frac{\mathbf{u} \cdot \hat{\mathbf{k}}}{c} \right)$$

where $\hat{\mathbf{k}} = \mathbf{k}/k$ is the unit propagation vector.

8.4 Wave Vector Scaling via Field-Mass Inertia Because the electromagnetic field carries mass-energy, the wave packet moving at velocity $\mathbf{u}$ carries an additional kinetic energy factor [Section 2.6, 7.2, 16.1]:

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$$

This scales the magnitude of the wave vector k (representing the momentum density of the wavefronts) relative to the source's intrinsic wave vector $k_0 = \omega_0/c$:

$$k = \gamma k_0 = \frac{k_0}{\sqrt{1 - \frac{u^2}{c^2}}}$$

8.5 Redoing the General 3D Doppler Shift Substituting this scaled wave vector k into the derived convective dispersion relation:

$$\omega_{\text{rec}} = c \left(\frac{k_0}{\sqrt{1 - \frac{u^2}{c^2}}} \right) \left(1 + \frac{\mathbf{u} \cdot \hat{\mathbf{k}}}{c} \right)$$

Since $ck_0 = \omega_0$ (where ω_0 is the intrinsic precessional frequency of the source):

$$\omega_{\text{rec}} = \omega_0 \frac{1 + \frac{\mathbf{u} \cdot \hat{\mathbf{k}}}{c}}{\sqrt{1 - \frac{u^2}{c^2}}}$$

Let the unit vector pointing from the stationary target to the source at the moment of emission be $\hat{\mathbf{r}}$. The wave propagates from the source to the target, so the unit propagation vector is $\hat{\mathbf{k}} = -\hat{\mathbf{r}}$. Substituting this:

$$\omega_{\text{rec}} = \omega_0 \frac{1 - \frac{\mathbf{u} \cdot \hat{\mathbf{r}}}{c}}{\sqrt{1 - \frac{u^2}{c^2}}}$$

Replacing the angular frequencies with standard frequencies ($\omega = 2\pi\nu$), and defining the radial component of the source's velocity along the line of propagation as $u_{\text{rad}} = \mathbf{u} \cdot \hat{\mathbf{r}}$:

$$\nu = \nu_0 \frac{1 - \frac{u_{\text{rad}}}{c}}{\sqrt{1 - \frac{u^2}{c^2}}}$$

This $Cl(4, 1, 1)$ field theory derivation demonstrates that the “relativistic” Doppler shift arises from the classical, convective bivector field equations and the physical mass-energy of electromagnetic waves moving inertially alongside their source.

8.6 Evaluation in Ordinary 3D Euclidean Space

In ordinary 3D Euclidean space $\mathbb{R}^3$, we represent the physical kinematics using standard Cartesian components.

Let the stationary target (the detector) be at the coordinate origin:

$$\mathbf{x}_{\text{rec}} = (0, 0, 0)$$

Let the moving transmission source have a constant velocity vector:

$$\mathbf{u} = (u_x, u_y, u_z)$$

The physical speed of the source is the Euclidean norm of its velocity vector:

$$u = |\mathbf{u}| = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

Let $\hat{\mathbf{r}}$ be the unit direction vector pointing from the target to the source at the moment of wave emission. In spherical coordinates:

$$\hat{\mathbf{r}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

The radial velocity component u_{rad} is evaluated as the standard Euclidean inner product:

$$u_{\text{rad}} = \mathbf{u} \cdot \hat{\mathbf{r}} = u_x \sin \theta \cos \phi + u_y \sin \theta \sin \phi + u_z \cos \theta$$

Substituting this into the general frequency relation:

$$\nu = \nu_0 \frac{1 - \frac{u_x \sin \theta \cos \phi + u_y \sin \theta \sin \phi + u_z \cos \theta}{c}}{\sqrt{1 - \frac{u_x^2 + u_y^2 + u_z^2}{c^2}}}$$

We evaluate this general Euclidean expression for three specific physical configurations:

8.6.1 Longitudinal Recession Case The transmission source moves directly away from the target along a straight line of sight. We align the coordinate system such that the motion and emission are on the x -axis. * The unit vector pointing toward the source is:

$$\hat{\mathbf{r}} = (1, 0, 0) \implies \theta = \frac{\pi}{2}, \phi = 0$$

* The velocity vector of the source is:

$$\mathbf{u} = (u, 0, 0)$$

* The radial velocity component is:

$$u_{\text{rad}} = \mathbf{u} \cdot \hat{\mathbf{r}} = u(1) + 0(0) + 0(0) = u$$

Substituting these values into the frequency relation:

$$\nu = \nu_0 \frac{1 - \frac{u}{c}}{\sqrt{1 - \frac{u^2}{c^2}}} = \nu_0 \frac{1 - \frac{u}{c}}{\sqrt{(1 - \frac{u}{c})(1 + \frac{u}{c})}} = \nu_0 \sqrt{\frac{1 - \frac{u}{c}}{1 + \frac{u}{c}}}$$

8.6.2 Longitudinal Approach Case The transmission source moves directly toward the target along the line of sight. * The unit vector pointing toward the source is:

$$\hat{\mathbf{r}} = (1, 0, 0) \implies \theta = \frac{\pi}{2}, \phi = 0$$

* The velocity vector of the source is:

$$\mathbf{u} = (-u, 0, 0)$$

* The radial velocity component is:

$$u_{\text{rad}} = \mathbf{u} \cdot \hat{\mathbf{r}} = -u(1) + 0(0) + 0(0) = -u$$

Substituting these values into the frequency relation:

$$\nu = \nu_0 \frac{1 + \frac{u}{c}}{\sqrt{1 - \frac{u^2}{c^2}}} = \nu_0 \frac{1 + \frac{u}{c}}{\sqrt{(1 - \frac{u}{c})(1 + \frac{u}{c})}} = \nu_0 \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}$$

8.6.3 Transverse Case The transmission source moves perpendicular to the line of sight at the moment of emission (such as at the point of closest approach).

* The unit vector pointing toward the source is:

$$\hat{\mathbf{r}} = (0, 1, 0) \implies \theta = \frac{\pi}{2}, \phi = \frac{\pi}{2}$$

* The velocity vector of the source is:

$$\mathbf{u} = (u, 0, 0)$$

* The radial velocity component is:

$$u_{\text{rad}} = \mathbf{u} \cdot \hat{\mathbf{r}} = u(0) + 0(1) + 0(0) = 0$$

Substituting these values into the frequency relation:

$$\nu = \nu_0 \frac{1 - 0}{\sqrt{1 - \frac{u^2}{c^2}}} = \frac{\nu_0}{\sqrt{1 - \frac{u^2}{c^2}}}$$

This completes the physical evaluation of the three-dimensional convective field equations, demonstrating that the longitudinal and transverse Doppler cases arise natively from ordinary Euclidean projections of the general $Cl(4, 1, 1)$ wave vector.